Report

Of

# PYTHON PROGRAMMING

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

25SCS1003005029

**TEJANSH VAIBHAV**

Batch: 2025-29

2025-29

## CANDIDATE’S DECLARATION

I, TEJANSH VAIBHAV, do hereby declare that the internship report titled “Python Programming” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: TEJANSH VAIBHAV

Roll No.: 25SCS1003005029

## ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank CodeAlpha for providing me the opportunity to participate in the Virtual Python Programming Internship and gain hands-on experience through practical assignments and projects. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Student Name: TEJANSH VAIBHAV

Roll No.: 25SCS1003005029

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# 3. INTERNSHIP COMPLETION CERTIFICATE

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Student ID:CA/DF1/182298

CERTIFICATE
OF COMPLETION

This Certificate Is Proudly Presented To

Tejansh Vaibhav

Was an active Participant at CodeAlpha Virtual Internship Program in Python Programming Internship with dedication and hard work. We really appreciate your efforts taken and wish you all the best for the further.
1st July 2026 to 30th July 2026

CEO & Founder

31st July 2026
Date of Issue

# 4. PROJECT DESCRIPTION

## 4.1 Introduction

This report presents a detailed account of the Virtual Python Programming Internship undertaken at CodeAlpha from 1st July 2026 to 30th July 2026. The internship provided practical exposure to Python programming through hands-on assignments and project development. The major work completed during the internship included a Basic Chatbot, a Stock Portfolio Tracker, and a Hangman Game. These projects helped strengthen programming fundamentals, problem-solving ability, input validation, modular programming, data handling, and file operations.

The internship was conducted completely online and focused on practical assignments and projects related to Python Programming.

## 4.2 Organization Profile

CodeAlpha provided the Virtual Python Programming Internship. The program gave practical exposure through programming assignments and projects and was conducted online. The internship documents confirm the role as Python Programming Intern and the duration from 1st July 2026 to 30th July 2026.

## 4.3 Problem Statement

The main problem addressed during the internship was to apply Python programming concepts to practical, user-interactive applications. Instead of limiting learning to theoretical exercises, the projects required implementation of input handling, conditions, loops, functions, data structures, validation, file operations, and user-friendly terminal output.

The three projects addressed different practical scenarios: conversational interaction through a chatbot, management and export of virtual investment data through a stock portfolio tracker, and interactive word guessing through a Hangman game.

## 4.4 Project Objectives

1. To strengthen Python programming fundamentals through practical implementation.
2. To develop modular and readable Python programs using functions and control structures.

3. To implement robust input validation and error handling.
4. To work with data structures and file input/output operations.
5. To develop interactive terminal-based applications.
6. To improve problem-solving, logical thinking, testing, and debugging skills.
7. To document and maintain projects using GitHub repositories.

**4.5 Scope of the Project**

The scope of the internship work covers three Python applications. The Basic Chatbot demonstrates continuous interaction, predefined responses, input validation, randomized greetings, and modular functions. The Stock Portfolio Tracker demonstrates portfolio creation, stock selection, quantity handling, formatted summaries, and export to TXT and CSV files. The Hangman Game demonstrates random word selection, input validation, game-state tracking, win/lose detection, ASCII-art stages, and replay functionality. The projects are terminal-based and use Python's standard library.

The project implementations are maintained as separate GitHub repositories for Basic Chatbot, Stock Portfolio Tracker, and Hangman Game.

**4.6 Technologies and Tools Used**

| Technology / Tool | Use | Projects |
|---|---|---|
| Python 3.x | Core programming language | All three |
| Python Standard Library | Built-in functionality such as random, csv and os | All three |
| Functions & Control Structures | Modular programming, loops and decision making | All three |
| Dictionaries & Data Structures | Data storage and retrieval | Stock Portfolio Tracker |
| File I/O | Saving portfolio summaries to TXT and CSV | Stock Portfolio Tracker |

| | | |
|---|---|---|
| GitHub | Code hosting and project documentation | All three |

### 4.7 System Architecture

The applications follow a simple terminal-based architecture: user input is collected through the command line, the program validates and processes the input, application logic is executed through modular functions and control structures, and the result is displayed in the terminal. The Stock Portfolio Tracker additionally persists generated portfolio summaries into TXT and CSV files.

Basic Chatbot: User Input → Input Validation → Response Matching → Response Output → Continue/Exit.

Stock Portfolio Tracker: Stock Selection → Quantity Validation → Portfolio Data → Calculations → Formatted Summary → TXT/CSV Export.

Hangman Game: Random Word Selection → Letter Input → Validation → Game-State Update → Win/Lose Check → Replay/Exit.

### 4.8 Methodology

The internship work followed a practical project-based methodology. Each project was developed by understanding the required functionality, implementing the core logic, validating user inputs, testing different cases, and documenting the final implementation. The projects were maintained as separate GitHub repositories.

| Project | Core Functionality | Key Concepts | Output |
|---|---|---|---|
| Basic Chatbot | Conversational terminal application | Loops, conditions, strings, functions, validation | Interactive chatbot |
| Stock Portfolio Tracker | Create and summarize virtual portfolio | Dictionaries, loops, validation, calculations, file I/O | Portfolio summary + TXT/CSV |

| | | | |
|---|---|---|---|
| Hangman Game | Word-guessing terminal game | Random selection, loops, validation, game state, functions | Interactive game |

**4.9 Expected Outcomes**

8. Improved understanding of Python syntax and programming fundamentals.
9. Ability to build interactive terminal-based applications.
10. Improved use of functions, loops, conditions, strings, and data structures.
11. Practical experience in validation and error handling.
12. Hands-on experience with file operations using TXT and CSV formats.
13. Improved project documentation and code organization through GitHub.
14. Greater confidence in developing small Python applications independently.

**4.10 Certificates of Completion and Communication Proof**

The Internship Completion Certificate confirms successful completion of the one-month Virtual Python Programming Internship at CodeAlpha from 1st July 2026 to 30th July 2026. The reference/student ID is CA/DF1/182298.

The Internship Offer Letter dated 30th June 2026 confirms selection as a Python Programming Intern at CodeAlpha and states that the internship would be conducted completely online with practical assignments and projects related to Python Programming.

The original-colour supporting documents are attached on the following pages.

# ORIGINAL INTERNSHIP OFFER LETTER

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# INTERNSHIP OFFER LETTER

**30th June 2026** **STUDENT ID: CA/DF1/182298**

**Name: Tejansh Vaibhav**

Welcome to CodeAlpha

Congratulations**!** We are delighted to make you an offer as **Python Programming Internship** offer letter is to confirm your selection **Python Programming Internship** at **CodeAlpha** and welcome you for the same effective from 1st July 2026 to 30th July 2026. Hope you will be doing well.

This Internship is observed by CodeAlpha as being a learning opportunity for you.

Your internship will embrace orientation and give emphasis on learning new skills with a deeper understanding of concepts through hands-on application of the knowledge you gained as an intern. You will acknowledge your obligation to perform all work allocated to you to the best of your ability within lawful and reasonable direction given to you.

We look forward to a worthwhile and fruitful association which will make you equipped for future projects.

Wishing you the most enjoyable and truly meaningful internship program experience.

**Sincerely,**

Swati Sri

**Swati Srivastava**
Founder & CEO

CODEALPHA SOFTWARE DEVELOPMENT

GOVERNMENT OF INDIA
MSME
MINISTRY OF MICRO, SMALL & MEDIUM ENTERPRISES

**Phone** +91 9336576683

**Email** services@codealpha.tech

**Address** Lucknow U.P

# LETTER OF RECOMMENDATION

CODE ALPHA

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**LinkedIn:**
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**Mobile No:**
+91 9336576683

# CODEALPHA

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## Letter Of Recommendation

### TO WHOM IT MAY CONCERN

This is to certify that Tejansh Vaibhav has successfully completed the Virtual Internship Program at CodeAlpha for a duration of one month from 1st July 2026 to 30th July 2026 as a Python Programming Internship.

He/She exhibited performance in this role and made a valuable contribution to our organization during the internship period. He/She has excellent analytical skills and He/She quickly acquired new skills and adeptly adapted to emerging technologies, demonstrating a high level of productivity.

He/She consistently displayed a willingness to provide assistance and fostered strong collaboration with fellow team members. His/Her dedication and skills make him/her a valuable asset to any prospective employer, and CodeAlpha wholeheartedly recommend him/her for any future endeavor.

REGARDS,

CA/DF1/182298

**STUDENT I'D**

SwatiSri

**FOUNDER & CEO**

CODEALPHA CODE ALPHA SOFTWARE DEVELOPMENT

31st July 2026

**DATE**

## 5. BIBLIOGRAPHY/REFERENCES


15. CodeAlpha, “Virtual Python Programming Internship,” Internship Offer Letter, 30 June 2026.
16. CodeAlpha, “Certificate of Internship – Python Programming,” issued 30 August 2026.
17. Tejansh Vaibhav, “CodeAlpha_BasicChatbot,” GitHub repository.
18. Tejansh Vaibhav, “CodeAlpha_StockPortfolioTracker,” GitHub repository.
19. Tejansh Vaibhav, “CodeAlpha_HangmanGame,” GitHub repository.

Repository links:

https://github.com/TejanshVaibhav07/CodeAlpha_BasicChatbot

https://github.com/TejanshVaibhav07/CodeAlpha_StockPortfolioTracker

https://github.com/TejanshVaibhav07/CodeAlpha_HangmanGame